Tutorial - 1

① $$L = \frac{m}{2}\frac{(\dot{\sigma}^2 + \dot{\tau}^2)}{\sigma^2 + \tau^2}$$

$$\therefore \frac{\partial L}{\partial \sigma} = -\frac{m\sigma}{(\sigma^2+\tau^2)^2}(\dot{\sigma}^2 + \dot{\tau}^2).$$

$$\therefore \frac{\partial L}{\partial \dot{\sigma}} = \frac{m\dot{\sigma}}{\sigma^2+\tau^2}$$

$$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\sigma}}\right) = \frac{m\ddot{\sigma}}{\sigma^2+\tau^2} - \frac{2m\dot{\sigma}^2\sigma}{(\sigma^2+\tau^2)^2} - \frac{2m\dot{\sigma}\tau\dot{\tau}}{(\sigma^2+\tau^2)^2}$$

$$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\sigma}}\right) - \frac{\partial L}{\partial \sigma} = 0$$

$$\Rightarrow \frac{m\ddot{\sigma}}{\sigma^2+\tau^2} - \frac{2m\dot{\sigma}}{(\sigma^2+\tau^2)^2}(\sigma\dot{\sigma} + \tau\dot{\tau}) + \frac{m\sigma}{(\sigma^2+\tau^2)^2}(\dot{\sigma}^2 + \dot{\tau}^2) = 0$$

$$\Rightarrow \ddot{\sigma}(\sigma^2+\tau^2) - 2\dot{\sigma}(\sigma\dot{\sigma} + \tau\dot{\tau}) + \sigma(\dot{\sigma}^2 + \dot{\tau}^2) = 0$$

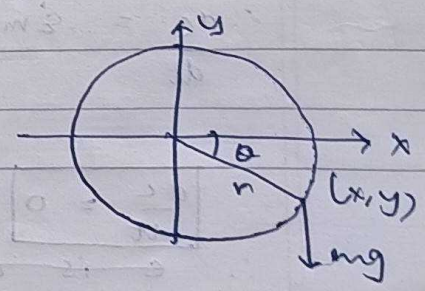
$$\Rightarrow \ddot{\sigma}(\sigma^2+\tau^2) - 2\dot{\sigma}\tau\dot{\tau} - \sigma\dot{\sigma}^2 \cancel{- 2\dot{\sigma}\dot{\tau}\tau} + \sigma\dot{\tau}^2 = 0$$

$$\Rightarrow \ddot{\sigma}(\sigma^2+\tau^2) + \sigma(\dot{\tau}^2 - \dot{\sigma}^2) - 2\dot{\sigma}\dot{\tau}\tau = 0. \quad \text{(Ans)}$$

Similarly: $\ddot{\tau}(\sigma^2+\tau^2) + \tau(\dot{\sigma}^2 - \dot{\tau}^2) - 2\dot{\tau}\dot{\sigma}\sigma = 0$ (Ans)

[$\because$ L is symmetric in σ & τ]

② GC: θ



$\therefore\ x = r\cos\theta \quad ; \quad y = r\sin\theta$ [We have considered $a = r$].

$\therefore\ \dot{x} = -r\sin\theta\,\dot{\theta} \quad ; \quad \dot{y} = r\cos\theta\,\dot{\theta}$

$\therefore\ v^2 = \dot{x}^2 + \dot{y}^2 = r^2\dot{\theta}^2$

$\therefore\ T = \frac{1}{2}mv^2 = \frac{1}{2}mr^2\dot{\theta}^2$

$\therefore\ V = +mgy = mgr\sin\theta$

$\therefore\ L = T - V = \frac{1}{2}mr^2\dot{\theta}^2 - mgr\sin\theta$ (Ans)